

Ghosh Exponent Theory

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Abstract

This paper develops the Ghosh Exponent Theory as an intrinsic mathematical framework centered on the exponent transformation

$$G_i(R) = R^{2/i}, \quad 0 \leq R \leq 1, \quad i \in \mathbb{N}, \quad i \geq 1. \quad (1)$$

The associated exact identity is

$$G_i(R)^i = R^2. \quad (2)$$

The theory develops the geometry of the transformation, its boundary behavior, exponent ordering, local sensitivity, pairwise exponent discrimination, discrete- R structures, exact intersections, finite-sample perturbation theory, and computational representations. Particular attention is given to the exceptional role of $i = 2$, for which the transformation is the identity, and to the exact relationship

$$R_i^{(S)} = R_{2,i}^{(\Delta)} = \left(\frac{2}{i}\right)^{i/(i-2)}, \quad i \geq 3. \quad (3)$$

The paper distinguishes structural mathematics from empirical inference. Since $G_i = R^{2/i}$ implies $G_i^i = R^2$ identically, the structural residual vanishes identically when G_i is defined from R ; consequently, that residual cannot by itself be used to select an exponent. Empirical model selection requires independently observed or estimated quantities. Applications to economics, finance, physics, chemistry, biology, psychology, psychiatry, philosophy, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, computer science, artificial intelligence, machine learning, and data science are presented as possible domains of application rather than as premises of the mathematical theory.

Contents

1	Introduction	3
2	Foundational Definitions	3
3	Elementary Geometry	4
4	Boundary Theory	4
5	Exponent Ordering	5
6	Sensitivity Theory	5
7	Pairwise Exponent Discrimination	6
8	Coordinates at the Discrimination Node	7
9	Adjacent Exponents	7
10	Sensitivity–Discrimination Intersection	8
11	The Discrete-R Family	8
12	Pairwise Ghosh Lattice	9
13	General Intersection Theory	9
14	A Non-Reference Exact Intersection	10

15	Multiplicative Structure of Intersections	10
16	Statistical Interpretation	11
17	Finite-Sample Perturbation Theory	11
18	Empirical Exponent Comparison	12
19	Algorithmic Formulation	12
20	Data Science	13
21	Artificial Intelligence and Machine Learning	13
22	Physics	13
23	Chemistry	14
24	Biology	14
25	Psychology and Psychiatry	14
26	Economics and Finance	14
27	Welfare Economics	15
28	Political Science, Geopolitics, Diplomacy, and International Relations	15
29	Warfare Economics and Strategic Capability	15
30	Philosophy	15
31	Religion and Metaphysical Interpretation	16
32	Systems Theory	16
33	Numerical Stability	16
34	Visualization	17
35	Unified Ghosh Statistical Architecture	17
36	Principal Theoretical Results	17
37	Ghosh Discrete- R Principle	18
38	Open Problems	18
39	Conclusion	19

List of Figures

1	Conceptual geometry of selected Ghosh exponent transformations. The dashed diagonal is the identity transformation $G_2 = R$. For $0 < R < 1$, the $i = 1$ transformation lies below the identity, while the $i = 3$ transformation lies above it.	17
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List of Tables

1	Initial members of the intrinsic sensitivity–discrimination family.	8
2	Selected pairwise discrimination nodes and corresponding Ghosh coordinates.	9
3	Intrinsic architecture of the Ghosh Exponent Theory.	17

1 Introduction

A bounded scalar state variable can be represented through a family of nonlinear exponent transformations. The Ghosh Exponent Theory begins with

$$G_i(R) = R^{2/i}, \quad 0 \leq R \leq 1, \quad i \in \mathbb{N}, \quad i \geq 1. \quad (4)$$

The corresponding exact power identity is

$$G_i(R)^i = R^2. \quad (5)$$

The central mathematical questions are:

1. What is the geometry of the transformation $R \mapsto G_i(R)$?
2. Which exponents are structurally exceptional?
3. How does the transformation respond to perturbations in R ?
4. At which values of R are competing exponent representations maximally separated?
5. Which discrete values of R arise intrinsically from these properties?
6. Where do sensitivity and discrimination structures intersect?
7. How should finite-sample observations be treated without confusing structural identities with empirical evidence?
8. How can the framework be generalized to multidimensional and computational settings?

The theory is deliberately developed from its own mathematical structure. Conventional statistical procedures may subsequently be used for empirical comparison, but they do not define the Ghosh objects themselves.

2 Foundational Definitions

Definition 2.1 (Ghosh exponent transformation). *For $R \in [0, 1]$ and $i \in \mathbb{N}$, $i \geq 1$, the i -th Ghosh exponent transformation is*

$$G_i(R) = R^{2/i}. \quad (6)$$

Definition 2.2 (Exact Ghosh identity). *The exact Ghosh identity associated with exponent i is*

$$G_i^i = R^2. \quad (7)$$

Definition 2.3 (Generalized residual representation). *When G_i and R are treated as independently measured or estimated quantities, define*

$$\varepsilon_i = R^2 - G_i^i. \quad (8)$$

The exact Ghosh relationship corresponds to

$$\varepsilon_i = 0. \quad (9)$$

Remark 2.4. *If G_i is defined directly by $G_i = R^{2/i}$, then $\varepsilon_i = 0$ identically. Consequently, the residual becomes informative for empirical testing only when G_i is independently observed, estimated, predicted, or otherwise supplied by a separate measurement process.*

Definition 2.5 (Ghosh sensitivity). *The local sensitivity of G_i with respect to R is*

$$S_i(R) = \frac{\partial G_i}{\partial R}. \quad (10)$$

Definition 2.6 (Pairwise Ghosh discrimination). *For $i < j$, define*

$$\Delta_{i,j}(R) = G_j(R) - G_i(R). \quad (11)$$

The pairwise discrimination node is the interior value of R at which this separation is maximized.

3 Elementary Geometry

For $0 < R < 1$,

$$\log G_i(R) = \frac{2}{i} \log R. \quad (12)$$

Consequently, the exponent appears linearly in logarithmic coordinates. Differentiating with respect to R gives

$$S_i(R) = \frac{2}{i} R^{2/i-1}. \quad (13)$$

The second derivative is

$$\frac{\partial^2 G_i}{\partial R^2} = \frac{2}{i} \left(\frac{2}{i} - 1 \right) R^{2/i-2}. \quad (14)$$

For $i = 2$,

$$G_2(R) = R, \quad (15)$$

and therefore

$$S_2(R) = 1. \quad (16)$$

Moreover,

$$\frac{\partial^2 G_2}{\partial R^2} = 0. \quad (17)$$

Thus $i = 2$ is structurally exceptional.

Theorem 3.1 (Identity exponent). *The unique positive exponent i for which*

$$G_i(R) = R \quad (18)$$

for every $R \in (0, 1)$ is

$$\boxed{i = 2}. \quad (19)$$

Proof. For $0 < R < 1$,

$$R^{2/i} = R \quad (20)$$

implies

$$\frac{2}{i} = 1. \quad (21)$$

Hence

$$i = 2. \quad (22)$$

□

□

Corollary 3.2. *For $0 < R < 1$,*

$$i < 2 \implies G_i(R) < R, \quad (23)$$

$$i = 2 \implies G_i(R) = R, \quad (24)$$

and

$$i > 2 \implies G_i(R) > R. \quad (25)$$

Proof. Since $0 < R < 1$, the function $a \mapsto R^a$ is strictly decreasing. The exponent $2/i$ is respectively greater than, equal to, or less than 1 according as $i < 2$, $i = 2$, or $i > 2$. □ □

4 Boundary Theory

At the lower boundary,

$$G_i(0) = 0 \quad (26)$$

for every positive integer i .

At the upper boundary,

$$G_i(1) = 1 \quad (27)$$

for every positive integer i .

Thus

$$R = 0 \implies G_i = 0 \quad \forall i, \quad (28)$$

and

$$R = 1 \implies G_i = 1 \quad \forall i. \quad (29)$$

Proposition 4.1 (Boundary degeneracy). *The exponent i cannot be distinguished from G_i alone at either $R = 0$ or $R = 1$.*

Proof. At each boundary every exponent produces the same transformed value. Hence the mapping from i to G_i is constant there. $\square$

The boundary therefore contains invariant states but no exponent discrimination.

5 Exponent Ordering

For $0 < R < 1$,

$$\log G_i = \frac{2 \log R}{i}. \quad (30)$$

Treating i temporarily as a positive real variable,

$$\frac{\partial \log G_i}{\partial i} = -\frac{2 \log R}{i^2}. \quad (31)$$

Since $\log R < 0$,

$$\frac{\partial \log G_i}{\partial i} > 0. \quad (32)$$

Therefore G_i is strictly increasing in i throughout the interior.
The logarithmic exponent sensitivity is

$$D_i(R) = \left| \frac{\partial \log G_i}{\partial i} \right| = \frac{2 |\log R|}{i^2}. \quad (33)$$

For fixed i ,

$$\frac{\partial D_i}{\partial R} = -\frac{2}{i^2 R} < 0. \quad (34)$$

Thus logarithmic sensitivity to the exponent increases as R decreases.

This quantity does not possess an interior maximum; its supremum on $(0, 1)$ is approached as $R \rightarrow 0^+$.

6 Sensitivity Theory

The sensitivity function is

$$S_i(R) = \frac{2}{i} R^{2/i-1}. \quad (35)$$

For $i = 2$,

$$S_2(R) = 1. \quad (36)$$

For $i > 2$,

$$\frac{2}{i} - 1 < 0, \quad (37)$$

so

$$S_i(R) \longrightarrow \infty \quad \text{as} \quad R \longrightarrow 0^+. \quad (38)$$

Definition 6.1 (Unit-sensitivity node). *For $i > 2$, the unit-sensitivity node $R_i^{(S)}$ is the unique interior solution of*

$$S_i(R) = 1. \quad (39)$$

Theorem 6.2 (Unit-sensitivity node). *For every integer $i > 2$,*

$$R_i^{(S)} = \left(\frac{2}{i} \right)^{i/(i-2)}. \quad (40)$$

Proof. The equation $S_i(R) = 1$ gives

$$\frac{2}{i} R^{2/i-1} = 1, \quad (41)$$

$$R^{(2-i)/i} = \frac{i}{2}. \quad (42)$$

Raising both sides to the power $i/(2-i)$ gives

$$R = \left(\frac{i}{2}\right)^{i/(2-i)} \quad (43)$$

$$= \left(\frac{2}{i}\right)^{i/(i-2)}. \quad (44)$$

□

□

The first members are

$$R_3^{(S)} = \left(\frac{2}{3}\right)^3 = \frac{8}{27}, \quad (45)$$

$$R_4^{(S)} = \left(\frac{2}{4}\right)^2 = \frac{1}{4}, \quad (46)$$

and

$$R_6^{(S)} = \left(\frac{2}{6}\right)^{3/2} = \frac{1}{3\sqrt{3}}. \quad (47)$$

At a unit-sensitivity node,

$$\Delta G_i = \Delta R + O((\Delta R)^2). \quad (48)$$

Thus the first-order transmission gain is exactly unity.

7 Pairwise Exponent Discrimination

For $i < j$, define

$$\Delta_{i,j}(R) = R^{2/j} - R^{2/i}. \quad (49)$$

Because $0 < R < 1$ and $2/j < 2/i$,

$$\Delta_{i,j}(R) > 0. \quad (50)$$

Differentiation gives

$$\frac{\partial \Delta_{i,j}}{\partial R} = \frac{2}{j} R^{2/j-1} - \frac{2}{i} R^{2/i-1}. \quad (51)$$

A stationary point therefore satisfies

$$\frac{1}{j} R^{2/j-1} = \frac{1}{i} R^{2/i-1}, \quad (52)$$

$$R^{2/i-2/j} = \frac{i}{j}. \quad (53)$$

Since

$$\frac{2}{i} - \frac{2}{j} = \frac{2(j-i)}{ij}, \quad (54)$$

we obtain the following theorem.

Theorem 7.1 (Pairwise discrimination node). *For integers $1 \leq i < j$, the unique interior maximum of*

$$\Delta_{i,j}(R) = R^{2/j} - R^{2/i} \quad (55)$$

occurs at

$$\boxed{R_{i,j}^{(\Delta)} = \left(\frac{i}{j}\right)^{ij/[2(j-i)]}}. \quad (56)$$

Proof. The stationary equation gives

$$R^{2(j-i)/(ij)} = \frac{i}{j}. \quad (57)$$

Because

$$0 < \frac{i}{j} < 1, \quad (58)$$

there is exactly one solution in $(0, 1)$. Furthermore,

$$\Delta_{i,j}(0) = 0 \quad (59)$$

and

$$\Delta_{i,j}(1) = 0, \quad (60)$$

while the separation is positive in the interior. Hence the unique interior stationary point is the global maximum. $\square$ $\square$

8 Coordinates at the Discrimination Node

Let

$$R_{i,j}^{(\Delta)} = \left(\frac{i}{j}\right)^{ij/[2(j-i)]}. \quad (61)$$

Then

$$G_i \left(R_{i,j}^{(\Delta)} \right) = \left(\frac{i}{j}\right)^{j/(j-i)}, \quad (62)$$

$$G_j \left(R_{i,j}^{(\Delta)} \right) = \left(\frac{i}{j}\right)^{i/(j-i)}. \quad (63)$$

Their ratio is

$$\frac{G_i \left(R_{i,j}^{(\Delta)} \right)}{G_j \left(R_{i,j}^{(\Delta)} \right)} = \left(\frac{i}{j}\right)^{(j-i)/(j-i)} \quad (64)$$

$$= \frac{i}{j}. \quad (65)$$

Hence

$$\boxed{\frac{G_i^*}{G_j^*} = \frac{i}{j}}. \quad (66)$$

The maximum separation is

$$\Delta_{i,j}^* = G_j^* - G_i^* \quad (67)$$

$$= \left(\frac{i}{j}\right)^{i/(j-i)} - \left(\frac{i}{j}\right)^{j/(j-i)} \quad (68)$$

$$= \frac{j-i}{j} \left(\frac{i}{j}\right)^{i/(j-i)}. \quad (69)$$

Therefore

$$\boxed{\Delta_{i,j}^* = \frac{j-i}{j} \left(\frac{i}{j}\right)^{i/(j-i)}}. \quad (70)$$

9 Adjacent Exponents

For

$$j = i + 1, \quad (71)$$

the discrimination node becomes

$$\boxed{R_{i,i+1}^{(\Delta)} = \left(\frac{i}{i+1}\right)^{i(i+1)/2}}. \quad (72)$$

The first cases are

$$R_{1,2}^{(\Delta)} = \frac{1}{2}, \quad (73)$$

$$R_{2,3}^{(\Delta)} = \frac{8}{27}, \quad (74)$$

and

$$R_{3,4}^{(\Delta)} = \frac{729}{4096}. \quad (75)$$

The corresponding maximum separations are

$$\Delta_{i,i+1}^* = \frac{1}{i+1} \left(\frac{i}{i+1} \right)^i. \quad (76)$$

Using

$$\left(1 - \frac{1}{i+1} \right)^i \rightarrow e^{-1}, \quad (77)$$

we obtain

$$\Delta_{i,i+1}^* \sim \frac{1}{e(i+1)}. \quad (78)$$

Hence adjacent high-order exponents become increasingly close in raw G -space.

10 Sensitivity–Discrimination Intersection

For $i > 2$,

$$R_i^{(S)} = \left(\frac{2}{i} \right)^{i/(i-2)}. \quad (79)$$

The pairwise discrimination node for the pair $(2, i)$ is

$$R_{2,i}^{(\Delta)} = \left(\frac{2}{i} \right)^{2i/[2(i-2)]} \quad (80)$$

$$= \left(\frac{2}{i} \right)^{i/(i-2)}. \quad (81)$$

Therefore:

Theorem 10.1 (Ghosh Sensitivity–Discrimination Intersection Theorem). *For every integer $i \geq 3$,*

$$\boxed{R_i^{(S)} = R_{2,i}^{(\Delta)} = \left(\frac{2}{i} \right)^{i/(i-2)}}. \quad (82)$$

Proof. The two expressions are algebraically identical. □ □

This establishes a structural role for the identity exponent $i = 2$: every unit-sensitivity node of a higher exponent is simultaneously the maximum-discrimination point between that exponent and the identity transformation.

11 The Discrete- R Family

The sensitivity–discrimination family is

$$\boxed{\mathcal{R}_{\text{SI}} = \left\{ \left(\frac{2}{i} \right)^{i/(i-2)} : i = 3, 4, 5, \dots \right\}}. \quad (83)$$

The first members are displayed in Table 1.

Table 1: Initial members of the intrinsic sensitivity–discrimination family.

i	$R_i^{(S)}$	$R_i^{(S)2}$	Property
3	8/27	64/729	Unit sensitivity of G_3
4	1/4	1/16	Unit sensitivity of G_4
5	$(2/5)^{5/3}$	$(2/5)^{10/3}$	Unit sensitivity of G_5
6	$1/(3\sqrt{3})$	1/27	Unit sensitivity of G_6
7	$(2/7)^{7/5}$	$(2/7)^{14/5}$	Unit sensitivity of G_7
8	$(1/4)^{4/3}$	$(1/4)^{8/3}$	Unit sensitivity of G_8

Taking logarithms gives

$$\log R_i^{(S)} = \frac{i}{i-2} \log \left(\frac{2}{i} \right). \quad (84)$$

Since

$$\log \left(\frac{2}{i} \right) \longrightarrow -\infty, \quad (85)$$

we obtain

$$R_i^{(S)} \longrightarrow 0 \quad \text{as} \quad i \longrightarrow \infty. \quad (86)$$

Thus the discrete sensitivity nodes accumulate at the lower boundary.

12 Pairwise Ghosh Lattice

The complete pairwise discrimination family is

$$\mathcal{R}_\Delta = \left\{ \left(\frac{i}{j} \right)^{ij/[2(j-i)]} : 1 \leq i < j \right\}. \quad (87)$$

Selected values are given in Table 2.

Table 2: Selected pairwise discrimination nodes and corresponding Ghosh coordinates.

Pair (i, j)	$R_{i,j}^{(\Delta)}$	G_i^*	G_j^*
(1, 2)	1/2	1/4	1/2
(2, 3)	8/27	8/27	4/9
(2, 4)	1/4	1/4	1/2
(3, 4)	729/4096	81/256	27/64
(3, 6)	1/8	1/4	1/2

The corrected entries in particular show that

$$R_{2,4}^{(\Delta)} = \left(\frac{2}{4} \right)^2 = \frac{1}{4}, \quad (88)$$

not 1/16, and

$$R_{3,6}^{(\Delta)} = \left(\frac{3}{6} \right)^3 = \frac{1}{8}, \quad (89)$$

not $3^{-9/2}$.

For the pair (3, 6),

$$G_3^* = \left(\frac{1}{8} \right)^{2/3} = \frac{1}{4}, \quad (90)$$

$$G_6^* = \left(\frac{1}{8} \right)^{1/3} = \frac{1}{2}. \quad (91)$$

13 General Intersection Theory

Suppose a sensitivity node coincides with a pairwise discrimination node:

$$R_n^{(S)} = R_{j,k}^{(\Delta)}. \quad (92)$$

Then

$$\left(\frac{2}{n} \right)^{n/(n-2)} = \left(\frac{j}{k} \right)^{jk/[2(k-j)]}. \quad (93)$$

Taking logarithms gives

$$\frac{n}{n-2} \log \left(\frac{n}{2} \right) = \frac{jk}{2(k-j)} \log \left(\frac{k}{j} \right). \quad (94)$$

Equivalently,

$$\boxed{\left(\frac{n}{2}\right)^{2n/(n-2)} = \left(\frac{k}{j}\right)^{jk/(k-j)}}. \quad (95)$$

The universal solution is

$$j = 2, \quad k = n. \quad (96)$$

Consequently,

$$R_n^{(S)} = R_{2,n}^{(\Delta)}. \quad (97)$$

14 A Non-Reference Exact Intersection

Consider

$$(n, j, k) = (18, 3, 9). \quad (98)$$

The sensitivity node is

$$R_{18}^{(S)} = \left(\frac{2}{18}\right)^{18/16} \quad (99)$$

$$= \left(\frac{1}{9}\right)^{9/8} \quad (100)$$

$$= 3^{-9/4}. \quad (101)$$

The discrimination node is

$$R_{3,9}^{(\Delta)} = \left(\frac{3}{9}\right)^{27/12} \quad (102)$$

$$= \left(\frac{1}{3}\right)^{9/4} \quad (103)$$

$$= 3^{-9/4}. \quad (104)$$

Hence

$$\boxed{R_{18}^{(S)} = R_{3,9}^{(\Delta)} = 3^{-9/4}}. \quad (105)$$

The associated coordinates are

$$G_3 = \left(3^{-9/4}\right)^{2/3} = 3^{-3/2}, \quad (106)$$

$$G_9 = \left(3^{-9/4}\right)^{2/9} = 3^{-1/2}. \quad (107)$$

Thus non-reference intersections exist beyond the universal family generated by the pair $(2, i)$.

15 Multiplicative Structure of Intersections

Suppose

$$\frac{n}{2} = q^a, \quad \frac{k}{j} = q^b, \quad (108)$$

where $q > 1$ and $a, b > 0$.

The intersection condition becomes

$$\frac{aq^a}{q^a - 1} = \frac{jbq^b}{2(q^b - 1)}. \quad (109)$$

Hence

$$\boxed{j = \frac{2aq^{a-b}(q^b - 1)}{b(q^a - 1)}}. \quad (110)$$

Once j is determined,

$$\boxed{k = jq^b}. \quad (111)$$

These equations provide a constructive route for searching for exact intersection triples, subject to the requirement that n, j, k satisfy the desired integer restrictions.

16 Statistical Interpretation

The Ghosh framework must distinguish between a mathematical transformation and an empirical measurement process.

If

$$G_i = R^{2/i}, \quad (112)$$

then

$$G_i^i = R^2 \quad (113)$$

holds identically.

Therefore,

$$\varepsilon_i = R^2 - G_i^i = 0 \quad (114)$$

for every admissible i .

This has an important consequence.

Proposition 16.1 (Structural residual degeneracy). *The residual*

$$\varepsilon_i = R^2 - G_i^i \quad (115)$$

cannot distinguish exponents when G_i is defined exactly as $R^{2/i}$.

Proof. Substitution gives

$$\varepsilon_i = R^2 - \left(R^{2/i}\right)^i \quad (116)$$

$$= R^2 - R^2 \quad (117)$$

$$= 0. \quad (118)$$

□

□

Thus the residual becomes statistically informative only if R and G_i are independent empirical quantities. For example, if observations satisfy

$$R_t^2 = G_{i,t}^i + \varepsilon_{i,t}, \quad (119)$$

where $G_{i,t}$ is independently measured, then the residual sequence can contain empirical information about the adequacy of exponent i .

This distinction prevents a tautological identity from being mistaken for an empirical statistical test.

17 Finite-Sample Perturbation Theory

Suppose the state variable is estimated from finite data:

$$\hat{R} = R + \eta. \quad (120)$$

The transformed estimator is

$$\hat{G}_i = \hat{R}^{2/i}. \quad (121)$$

A Taylor expansion gives

$$\hat{G}_i = G_i + S_i(R)\eta + \frac{1}{2} \frac{\partial^2 G_i}{\partial R^2} \eta^2 + O(\eta^3). \quad (122)$$

Therefore,

$$\hat{G}_i - G_i = S_i(R)\eta + O(\eta^2). \quad (123)$$

At the unit-sensitivity node,

$$S_i(R_i^{(S)}) = 1, \quad (124)$$

and hence

$$\hat{G}_i - G_i = \eta + O(\eta^2). \quad (125)$$

Thus the unit-sensitivity node provides a natural first-order calibration point for propagation of state uncertainty into the transformed coordinate.

18 Empirical Exponent Comparison

If G_i is independently observed or estimated, an empirical structural loss can be defined by

$$\mathcal{L}(i) = \frac{1}{N} \sum_{t=1}^N |R_t^2 - G_{i,t}^i|^p, \quad p \geq 1. \quad (126)$$

An empirical exponent estimate may then be defined as

$$\hat{i} = \arg \min_{i \in \mathcal{I}_0} \mathcal{L}(i), \quad (127)$$

where $\mathcal{I}_0$ is the candidate exponent set.

This criterion is meaningful precisely because $G_{i,t}$ is not assumed to be generated mechanically from R_t .

For repeated samples indexed by b ,

$$\hat{i}^{(b)} = \arg \min_{i \in \mathcal{I}_0} \mathcal{L}^{(b)}(i). \quad (128)$$

The empirical frequency with which exponent i is selected is

$$\hat{\pi}_i = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \{ \hat{i}^{(b)} = i \}. \quad (129)$$

This is a finite-sample stability statistic rather than a claim that a particular conventional inferential distribution is required.

19 Algorithmic Formulation

The basic structural computation can be written as follows.

INPUT:

```
state R
candidate exponent set I
candidate exponent pairs P
```

FOR each i in I :

```
compute  $G_i = R^{(2/i)}$ 
compute  $S_i = (2/i) R^{(2/i - 1)}$ 
```

FOR each pair (i, j) in P with $i < j$:

```
compute  $R_{\text{delta}} = (i/j)^{(i*j/(2*(j-i)))}$ 
compute  $G_{i\_star} = R_{\text{delta}}^{(2/i)}$ 
compute  $G_{j\_star} = R_{\text{delta}}^{(2/j)}$ 
```

FOR each $i > 2$:

```
compute  $R_{\text{sensitivity}} = (2/i)^{(i/(i-2))}$ 
```

RETURN:

```
Ghosh coordinates
sensitivities
discrimination nodes
sensitivity nodes
exact intersections
```

The computational complexity of evaluating a finite candidate set is linear in the number of candidate exponents and candidate exponent pairs, apart from the numerical cost of exponentiation.

20 Data Science

For a data vector

$$\mathbf{R} = (R_1, \dots, R_N), \quad (130)$$

the exponent transformation produces

$$\mathbf{G}_i = (R_1^{2/i}, \dots, R_N^{2/i}). \quad (131)$$

For a finite collection of exponents,

$$\mathcal{I}_0 = \{i_1, \dots, i_m\}, \quad (132)$$

one obtains the feature representation

$$\Phi(\mathbf{R}) = [\mathbf{G}_{i_1}, \mathbf{G}_{i_2}, \dots, \mathbf{G}_{i_m}]. \quad (133)$$

This produces an exponent-indexed nonlinear representation of the original data.

21 Artificial Intelligence and Machine Learning

A machine-learning system can use the family

$$G_i(R) = R^{2/i} \quad (134)$$

as a deterministic collection of nonlinear feature transformations.

A multi-exponent representation is

$$\mathbf{G}(R) = (R^{2/i_1}, \dots, R^{2/i_m}). \quad (135)$$

A possible structural loss is

$$\mathcal{L}_{\text{Ghosh}} = \sum_{t=1}^N \sum_{i \in \mathcal{I}_0} w_i |R_t^2 - G_{i,t}^i|^p, \quad (136)$$

where $w_i \geq 0$.

However, if

$$G_{i,t} = R_t^{2/i}, \quad (137)$$

then every term vanishes identically. Therefore, a learning problem requires independently generated predictions or measurements if the loss is to discriminate among exponents.

This observation is fundamental to avoiding circular model construction.

22 Physics

Suppose R is a normalized physical state satisfying

$$0 \leq R \leq 1. \quad (138)$$

The Ghosh transformation gives

$$G_i = R^{2/i}. \quad (139)$$

The sensitivity

$$S_i(R) = \frac{2}{i} R^{2/i-1} \quad (140)$$

then quantifies local propagation of changes in the normalized state.

The theory is agnostic regarding whether R represents normalized amplitude, intensity, occupancy, probability, or another dimensionless physical state.

23 Chemistry

A normalized chemical state may be represented by

$$R \in [0, 1]. \quad (141)$$

Potential examples include normalized activity, reaction progress, fractional occupancy, or another bounded state variable.

The Ghosh transformation is then

$$G_i = R^{2/i}. \quad (142)$$

Any physical or chemical interpretation must be established independently; the mathematical transformation does not itself imply a chemical mechanism.

24 Biology

A bounded biological state can similarly be represented through

$$G_i = R^{2/i}. \quad (143)$$

Possible applications include normalized population states, fractional activation, occupancy, response, or other bounded quantities.

The mathematical questions remain invariant:

$$\text{How does the exponent alter sensitivity?} \quad (144)$$

and

$$\text{Where are competing exponent representations maximally separated?} \quad (145)$$

25 Psychology and Psychiatry

Psychological measurements often involve normalized scores or bounded response variables.

If a normalized state R is theoretically justified, the Ghosh family supplies

$$G_i = R^{2/i}. \quad (146)$$

The sensitivity function provides a mathematical measure of local amplification:

$$S_i(R) = \frac{2}{i} R^{2/i-1}. \quad (147)$$

No clinical or psychiatric conclusion follows from this transformation alone. Any clinical application requires independent empirical validation.

26 Economics and Finance

Let R represent a normalized economic or financial state.

The Ghosh coordinate is

$$G_i = R^{2/i}. \quad (148)$$

Potential applications include normalized measures of risk, return magnitude, liquidity, financial stability, credit conditions, or other bounded variables.

The sensitivity is

$$S_i(R) = \frac{2}{i} R^{2/i-1}. \quad (149)$$

For $i > 2$,

$$G_i > R \quad (150)$$

in the interior, whereas

$$i < 2 \implies G_i < R. \quad (151)$$

The exponent $i = 2$ therefore provides an identity benchmark.

27 Welfare Economics

Let R denote a normalized welfare state:

$$0 \leq R \leq 1. \quad (152)$$

The exponent family is

$$G_i = R^{2/i}. \quad (153)$$

The pairwise separation is

$$\Delta_{i,j}(R) = R^{2/j} - R^{2/i}. \quad (154)$$

The value

$$R_{i,j}^{(\Delta)} = \left(\frac{i}{j}\right)^{ij/[2(j-i)]} \quad (155)$$

identifies the point of maximum coordinate separation between the two transformations.

This is a mathematical result and does not by itself constitute a welfare criterion.

28 Political Science, Geopolitics, Diplomacy, and International Relations

Suppose R represents a normalized political, diplomatic, geopolitical, or strategic state.

The transformation

$$G_i = R^{2/i} \quad (156)$$

generates a family of nonlinear state representations.

The identity exponent satisfies

$$G_2 = R. \quad (157)$$

Higher exponents satisfy

$$G_i > R \quad (158)$$

in the interior, while lower exponents satisfy

$$G_i < R. \quad (159)$$

The exponent may therefore be interpreted in an application as a parameter governing the curvature of a representation, provided that the application supplies an independent theoretical justification for the interpretation.

29 Warfare Economics and Strategic Capability

Suppose R is a normalized strategic capability variable.

Then

$$G_i = R^{2/i} \quad (160)$$

provides a nonlinear capability coordinate.

For competing representations $i < j$, maximum coordinate separation occurs at

$$R_{i,j}^{(\Delta)} = \left(\frac{i}{j}\right)^{ij/[2(j-i)]}. \quad (161)$$

The mathematical result can therefore identify states at which two strategic capability transformations are maximally separated, without asserting that either representation is empirically correct.

30 Philosophy

The theory distinguishes between an underlying state and a representation of that state.

The state is

$$R, \quad (162)$$

whereas

$$G_i = R^{2/i} \quad (163)$$

is an exponent-dependent representation.

The identity exponent gives

$$G_2 = R. \quad (164)$$

Thus $i = 2$ is the reference representation in which transformed coordinate and underlying state coincide. This provides a mathematical basis for distinguishing an object from its representation.

31 Religion and Metaphysical Interpretation

The boundary states satisfy

$$G_i(0) = 0 \quad (165)$$

and

$$G_i(1) = 1 \quad (166)$$

for every positive integer exponent.

The identity exponent satisfies

$$G_2(R) = R. \quad (167)$$

These properties can support philosophical or metaphysical interpretations concerning invariance, transformation, unity, and reference. Such interpretations remain analogical rather than mathematical consequences.

32 Systems Theory

Define

$$T_i(R) = R^{2/i}. \quad (168)$$

Composition gives

$$T_j(T_i(R)) = \left(R^{2/i}\right)^{2/j} \quad (169)$$

$$= R^{4/(ij)}. \quad (170)$$

For this composition to equal another Ghosh transformation

$$T_k(R) = R^{2/k}, \quad (171)$$

we require

$$\frac{4}{ij} = \frac{2}{k}. \quad (172)$$

Hence

$$\boxed{k = \frac{ij}{2}}. \quad (173)$$

Thus composition is closed within the family whenever $ij/2$ is an admissible exponent.

33 Numerical Stability

The transformation can be evaluated through logarithms:

$$G_i = \exp\left(\frac{2}{i} \log R\right). \quad (174)$$

Likewise,

$$R_{i,j}^{(\Delta)} = \exp\left[\frac{ij}{2(j-i)} \log\left(\frac{i}{j}\right)\right]. \quad (175)$$

The logarithmic representation is particularly useful when R is very close to zero or when the exponent indices are large.

For numerical calculations, underflow and loss of precision should be considered explicitly.

34 Visualization

Figure 1 provides a conceptual representation of the exponent family.

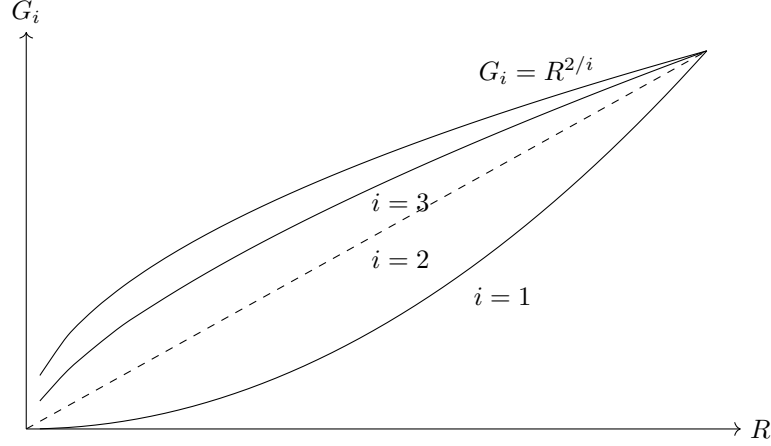


Figure 1: Conceptual geometry of selected Ghosh exponent transformations. The dashed diagonal is the identity transformation $G_2 = R$. For $0 < R < 1$, the $i = 1$ transformation lies below the identity, while the $i = 3$ transformation lies above it.

35 Unified Ghosh Statistical Architecture

The mathematical architecture is summarized in Table 3.

Table 3: Intrinsic architecture of the Ghosh Exponent Theory.

Layer	Mathematical object
State	$R \in [0, 1]$
Exponent	$i \in \mathbb{N}, i \geq 1$
Transformation	$G_i = R^{2/i}$
Exact identity	$G_i^i = R^2$
Independent-observation residual	$\varepsilon_i = R^2 - G_i^i$
Sensitivity	$S_i(R) = \frac{2}{i} R^{2/i-1}$
Unit-sensitivity node	$R_i^{(S)} = (2/i)^{i/(i-2)}$ for $i > 2$
Pairwise discrimination	$\Delta_{i,j} = G_j - G_i$
Discrimination node	$R_{i,j}^{(\Delta)} = (i/j)^{ij/[2(j-i)]}$
Reference exponent	$i = 2$
Universal intersection	$R_i^{(S)} = R_{2,i}^{(\Delta)}$
Finite-sample perturbation	$\hat{G}_i - G_i = S_i(R)\eta + O(\eta^2)$

36 Principal Theoretical Results

The principal results are:

1. The Ghosh transformation is

$$G_i = R^{2/i}. \quad (176)$$

2. Its exact power identity is

$$G_i^i = R^2. \quad (177)$$

3. The boundaries are invariant:

$$G_i(0) = 0, \quad G_i(1) = 1. \quad (178)$$

4. The exponent $i = 2$ is the unique interior identity exponent:

$$G_2 = R. \quad (179)$$

5. For $i > 2$, the unit-sensitivity node is

$$R_i^{(S)} = \left(\frac{2}{i}\right)^{i/(i-2)}. \quad (180)$$

6. For $i < j$, the pairwise discrimination node is

$$R_{i,j}^{(\Delta)} = \left(\frac{i}{j}\right)^{ij/[2(j-i)]}. \quad (181)$$

7. The sensitivity and identity-discrimination structures intersect exactly:

$$R_i^{(S)} = R_{2,i}^{(\Delta)}. \quad (182)$$

8. The maximum adjacent-exponent separation satisfies

$$\Delta_{i,i+1}^* \sim \frac{1}{e(i+1)}. \quad (183)$$

9. The adjacent discrimination nodes satisfy

$$R_{i,i+1}^{(\Delta)} \rightarrow 0. \quad (184)$$

10. The structural residual

$$R^2 - G_i^i \quad (185)$$

is identically zero if G_i is defined from R , and therefore cannot itself constitute an empirical exponent-selection statistic.

37 Ghosh Discrete- R Principle

The preceding results motivate the following principle.

Definition 37.1 (Ghosh Discrete- R Principle). *A value of R is structurally distinguished within the Ghosh framework when it is generated by an intrinsic property of the transformation, such as sensitivity, discrimination, invariance, or an exact intersection of such properties.*

This principle excludes arbitrary numerical selection.

The strongest currently derived single-index family is

$$\mathcal{R}_{\text{SI}} = \left\{ \left(\frac{2}{i}\right)^{i/(i-2)} : i \geq 3 \right\}. \quad (186)$$

The broader pairwise family is

$$\mathcal{R}_{\Delta} = \left\{ \left(\frac{i}{j}\right)^{ij/[2(j-i)]} : 1 \leq i < j \right\}. \quad (187)$$

The exact intersections between these families form a further arithmetic structure.

38 Open Problems

The following problems remain mathematically well-defined.

1. Characterize all integer triples (n, j, k) satisfying

$$R_n^{(S)} = R_{j,k}^{(\Delta)}. \quad (188)$$

2. Determine whether all non-reference intersections admit a complete characterization through multiplicative dependence among the rational ratios $n/2$ and k/j .

3. Characterize the complete intersection set

$$\mathcal{I} = \left\{ R : R_n^{(S)} = R_{j,k}^{(\Delta)} \text{ for admissible indices} \right\}. \quad (189)$$

4. Extend the sensitivity analysis to higher derivatives.

5. Develop the complete finite-sample error-propagation theory for

$$\widehat{G}_i = \widehat{R}^{2/i}. \quad (190)$$

6. Determine additional discrete- R families generated by curvature, higher-order sensitivity, or other intrinsic geometric criteria.

7. Characterize intersections among sensitivity, discrimination, curvature, and higher-order stability nodes.

8. Develop multidimensional Ghosh transformations of the form

$$G_{\mathbf{i}}(\mathbf{R}) = \prod_{r=1}^m R_r^{2/i_r}. \quad (191)$$

9. Construct empirical Ghosh models in which R and G_i arise from independent observations or independently specified mechanisms.

10. Determine the conditions under which a Ghosh exponent can be statistically identified from noisy observations.

39 Conclusion

The Ghosh Exponent Theory begins with the transformation

$$G_i(R) = R^{2/i}. \quad (192)$$

Despite its simplicity, this transformation generates a substantial intrinsic structure.

The boundary states

$$R = 0 \quad \text{and} \quad R = 1 \quad (193)$$

are invariant but exponent-degenerate.

The exponent

$$i = 2 \quad (194)$$

is uniquely distinguished by

$$G_2(R) = R. \quad (195)$$

For every $i > 2$, the unit-sensitivity node is

$$R_i^{(S)} = \left(\frac{2}{i} \right)^{i/(i-2)}. \quad (196)$$

For every $i < j$, the pairwise discrimination node is

$$R_{i,j}^{(\Delta)} = \left(\frac{i}{j} \right)^{ij/[2(j-i)]}. \quad (197)$$

The principal intersection is

$$\boxed{R_i^{(S)} = R_{2,i}^{(\Delta)}} \quad (198)$$

Consequently, the unit-sensitivity point of G_i is precisely the point at which G_i is maximally separated from the identity coordinate G_2 .

The resulting discrete- R structure is therefore generated by the internal geometry of the exponent family rather than by externally imposed statistical conventions.

At the same time, the theory requires a careful distinction between mathematics and empirical inference. Because

$$G_i = R^{2/i} \quad (199)$$

implies

$$G_i^i = R^2 \quad (200)$$

identically, the structural residual is zero by construction. Empirical identification of an exponent therefore requires an independently observed or estimated G_i , an independently specified model, or another non-tautological source of variation.

The central mathematical structure can therefore be stated compactly as

$$\boxed{\text{Ghosh transformation} \longrightarrow \text{sensitivity} \longrightarrow \text{pairwise discrimination} \longrightarrow \text{exact intersections} \longrightarrow \text{discrete R-structure}} \quad (201)$$

The framework is sufficiently general to admit applications across economics, finance, physics, chemistry, biology, psychology, psychiatry, philosophy, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, computer science, artificial intelligence, machine learning, data science, and other disciplines. In each application, however, the interpretation of R and i must be established independently of the mathematical identities themselves.

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Glossary

Ghosh exponent

The positive integer i indexing the transformation $G_i(R) = R^{2/i}$.

Ghosh exponent transformation

The map $R \mapsto R^{2/i}$.

Ghosh coordinate

The transformed quantity G_i .

Exact Ghosh identity

The identity $G_i^i = R^2$.

Structural residual

The quantity $\varepsilon_i = R^2 - G_i^i$ when R and G_i are treated as independently observed or estimated.

Identity exponent

The unique exponent $i = 2$ for which $G_i = R$ throughout the interior.

Boundary degeneracy

The fact that all positive exponents yield $G_i = 0$ at $R = 0$ and $G_i = 1$ at $R = 1$.

Sensitivity

The derivative $S_i(R) = \partial G_i / \partial R$.

Unit-sensitivity node

The interior value of R for which $S_i(R) = 1$.

Pairwise discrimination

The separation $G_j - G_i$ between two exponent representations.

Discrimination node

The value of R maximizing pairwise discrimination.

Sensitivity–discrimination intersection

A value of R that simultaneously satisfies a sensitivity criterion and a pairwise discrimination criterion.

Discrete- R family

A set of R -values generated by intrinsic mathematical properties of the Ghosh transformation.

Reference coordinate

The identity coordinate $G_2 = R$.

Exponent lattice

The collection of pairwise relationships generated by integer exponent pairs.

Finite-sample Ghosh theory

The analysis of how estimates of R propagate through the nonlinear Ghosh transformation.

Structural identification

Identification based on independently observed or theoretically specified quantities rather than tautological substitution.

Multiplicative dependence

A relationship in which positive rational quantities can be represented as powers of a common positive base.

High-exponent compression

The asymptotic reduction in absolute separation between adjacent high-order exponent transformations.

The End